

Adrian Engel Rodriguez

Biophysicist | Berlin, Germany – Madrid, Spain | (+49) 157 87 607 286

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Professional Summary

Biophysics master's student with a strong analytical and quantitative foundation and developing expertise in data science and financial analysis. Experienced in modeling complex systems and extracting insights from data using computational methods. Aspiring to work at the intersection of science, technology, and finance, applying data-driven analysis to support technical innovation and strategic financial decision-making.

Key Skills

- **Languages:** German (native tongue), Spanish (native tongue), English (advanced – C1/C2)
- **Microsoft:** Excel, Word, PowerPoint, Teams, OneNote, Outlook
- **Practical lab experience:** Physics, Biophysics, Biochemistry, Physical Chemistry, Cell Biology, Genetics, Animal Physiology, AFM, FTIR, Deep Learning, Machine Learning, Artificial Intelligence, Neurobiophysics (NEURON), Single Molecule Spectroscopy
- **Programming:** APIs, Bash (Linux), C, C++, CMD/Powershell (Windows), Data scraping, FIJI, Git, GitHub, HTML, ImageJ, Java, JavaScript, LaTeX, MATLAB, NEURON, Notepad++ (NPP/npp), Python, R, SQL, Version Control, Visual Studio Code (VS Code)
- **Python:** Google Colab, Jupyter Notebook, Matplotlib, NumPy, Pandas, PyTorch, SciPy, Seaborn

Work Experience

Student Office Worker – Austrian Economics & Classical Liberalism | April 2026 – Present

Friedrich August von Hayek-Gesellschaft e.V., Berlin, Germany

- Layout, design and production: Magazine "hayekforum" (Adobe InDesign, Canvas, AI tools)
- Social media content management (Instagram, LinkedIn, X)
- Organizing and assisting conferences, forums, meetings and summits

Student Study Advisor | August 2025 – Present

Department of Biology, Faculty of Life Sciences, Humboldt-Universität zu Berlin, Germany

- Tasked with developing study plans and schedules for students
- Answering organizational questions and referring students to the appropriate counselors
- Planning and organizing university events for incoming students
- Solving student's problems with the university bureaucracy, e.g. the examination office

Student Researcher – Study Project and Bachelor Thesis | May 2024 – October 2025

AG Romanczuk, Institute for Theoretical Biology, Humboldt-Universität zu Berlin, Germany

Role of Heterogeneity in the Self-Organized Collective Movement of a Simple Multicellular Organism

- Computational modeling of collective motion for a system of self-propelled particles
- Numerical simulations in Python and C
- Statistical/quantitative analysis (order parameter, susceptibility, phase transition)
- Data processing & visualization

Student Researcher – Voluntary Summer Internship | August 2023 – November 2023

Roderer Lab, Leibniz-Forschungsinstitut für Molekulare Pharmakologie, Berlin, Germany

- 3D reconstruction of a protein (MORN2) from single particle cryo-EM data
- Software and tools: cryoSPARC, PyMOL, UCSF Chimera, Topaz, AlphaFold
- Laboratory documentation, scientific research, critical thinking, group discussion, presenting, public speaking, problem-solving, team work

Mentor and Tutor for First Year Students | October 2021 – February 2022

Mentoring-Tutoring-Program, Humboldt-Universität zu Berlin, Germany

- Mentoring for soft skill development (Communication, Teamwork)
- Extramodular tutoring for introductory math, physics, chemistry and biology courses

Education

Master of Science in Biophysics, Humboldt-Universität zu Berlin, Germany | October 2025 – Present

- **Compulsory courses:** Statistical Physics and Quantum Mechanics
- **Compulsory elective courses:** Molecular, Cellular and Medical Biophysics, Neurobiophysics, Optobiology, Single Molecule Spectroscopy, Systems Biology
- **Elective courses:** Business Analytics & Data Science, Econometric Methods, Multivariate Statistical Analysis, Statistical Programming Languages, Time Series Analysis

Bachelor of Arts in Catholic Theology (digital), Katholische Universität Eichstätt-Ingolstadt & Universität Passau, Bavaria, Germany | October 2025 – Present

- Ancient Languages: Hebrew, Koine Greek, Latin
- Biblical Theology and Hermeneutics
- Bibliographical Research and Scientific Writing
- Historical Criticism and History of the Church
- Philosophical and Theological Thinking

Course in Multimarket Investing (online), Universidad de las Hespérides, Canary Islands, Spain | April 2025 – March 2026

- Personal finance & risk management
- Variable & fixed income investments, financial derivatives
- Business management, taxation and private equity
- Real estate investments, raw materials, geopolitical analysis
- Bitcoin and other crypto currencies

Bachelor of Science in Biophysics, Humboldt-Universität zu Berlin, Germany | October 2020 – October 2025

- **Compulsory courses:** Linear Algebra, Calculus, Statistics, Experimental Physics, Cell Biology, General and Inorganic Chemistry, Organic Chemistry, Physical Chemistry, Biochemistry and Microbiology, Experimental and Theoretical Biophysics, Programming and Bioinformatics
- **Compulsory elective courses:** Advanced Experimental and Theoretical Biophysics
- **Elective courses:** Cellular Biophysics, Systems Biology, Image Processing and Analysis, Bioinformatics, English C2 - Critical Writing, Mentoring-Tutoring-Program
- **Bachelor Thesis:** *Role of Heterogeneity in the Self-Organized Collective Movement of a Simple Multicellular Organism* - Criticality, Statistical Physics, Theoretical Biophysics

Erasmus Exchange, Universidad Autónoma de Madrid, Spain | September 2022 – July 2023

- Bachelor in Biology, Bachelor in Physics
- Master in Physics of Condensed Matter and Biological Systems

MINTgrün, Technische Universität Berlin, Germany | October 2018 – September 2019

- STEM field orientation program, 2 semesters

German High School, Friedrich-Engels-Gymnasium, Berlin, Germany | September 2010 – June 2018

- German-Spanish bilingual high school diploma, Grade: 1.6 (Scale: 1.0 = best, 4.0 = pass)

Professional Development

- Hayek-Gesellschaft
 - Akademie der Freiheit 2025 (Scholarship for a summer academy)
 - Juniorenkreis Politik & Juniorenkreis Wissenschaft (Youth seminars)

- Austrian Institute: Austrian Academy 2025 (Scholarship for a summer academy)

Projects

- Pokémon Rom Hacking
 - Source code manipulation of Pokémon video games for the Gameboy
 - Game design: Feature innovation, story-telling

Volunteering

- Spanish Red Cross: Summer activities supervisor for children at risk of social exclusion

Certifications

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